

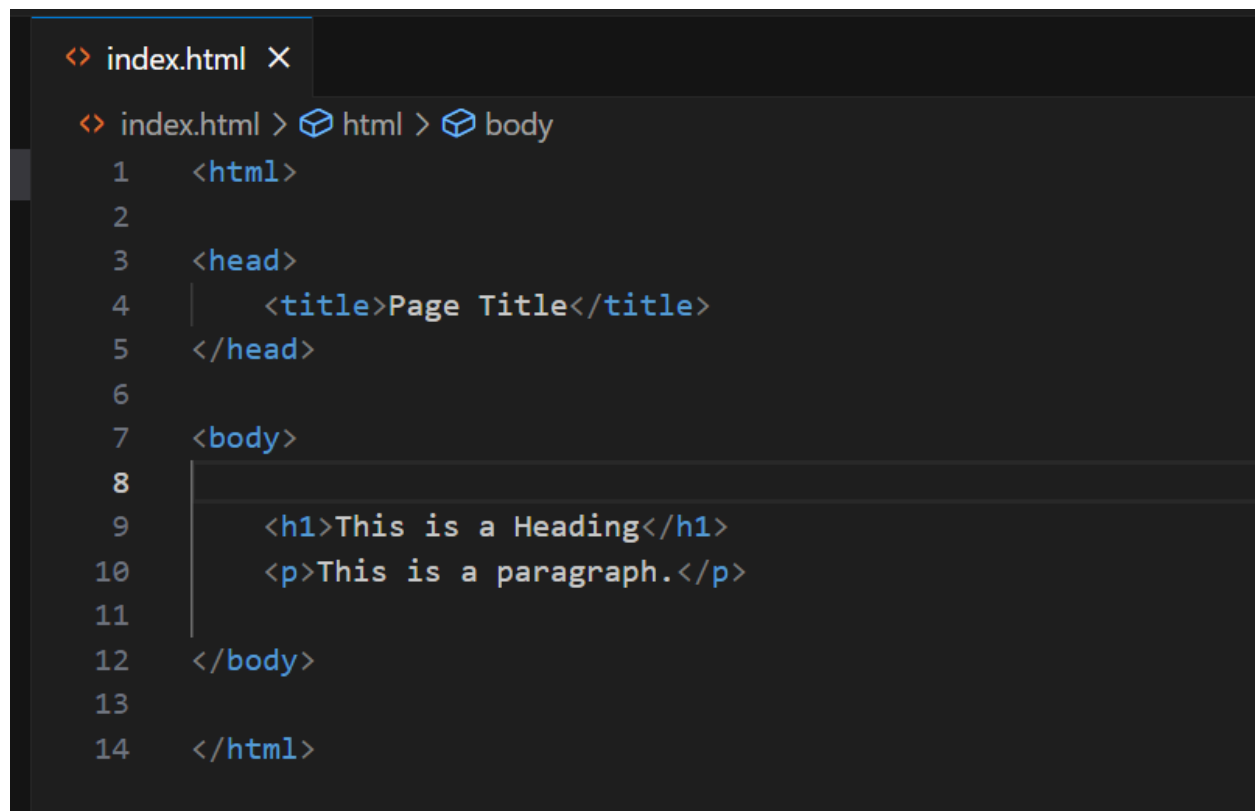
## HTML Basics

HTML is the standard markup language for Web pages.

HTML stands for Hyper Text Markup Language

With HTML you can create your own Website.

Below code shows the basic example and syntax of HTML

A screenshot of a code editor with a dark theme. At the top, there is a tab labeled 'index.html' with a close button. Below the tab, a breadcrumb navigation shows 'index.html > html > body'. The main code area displays an HTML document structure with line numbers on the left. The code starts with a <html> tag, followed by a <head> section containing a <title>Page Title</title> tag. The </head> tag is closed, followed by an opening <body> tag. Inside the body, there is an <h1>This is a Heading</h1> tag and a <p>This is a paragraph.</p> tag. The </body> tag is closed, and finally, the </html> tag is closed at the bottom.

```
1  <html>
2
3  <head>
4    <title>Page Title</title>
5  </head>
6
7  <body>
8
9    <h1>This is a Heading</h1>
10   <p>This is a paragraph.</p>
11
12 </body>
13
14 </html>
```

## HTML Elements

An HTML element is defined by a start tag, some content, and an end tag.

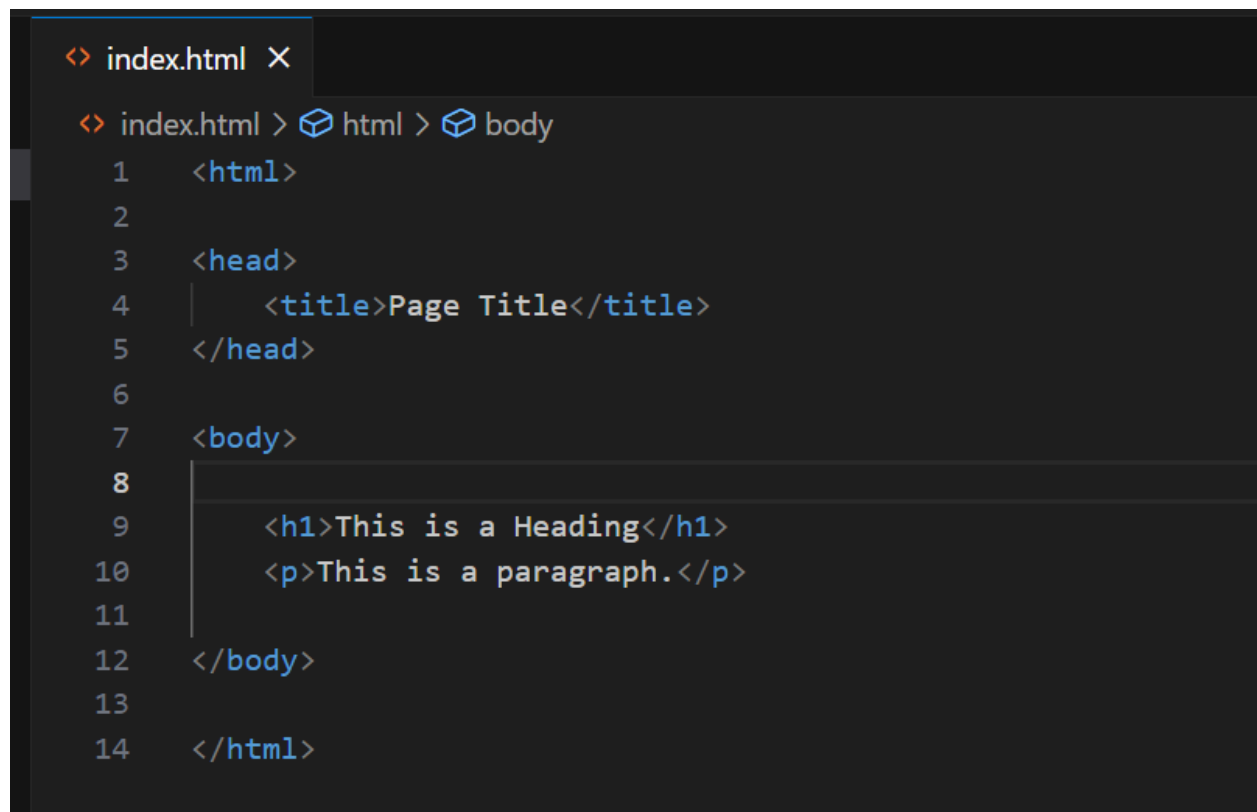
The HTML element is everything from the start tag to the end tag:

`<tagname>Content goes here...</tagname>`

Examples of some HTML elements:

`<h1>My First Heading</h1>`

`<p>My first paragraph.</p>`

A screenshot of a code editor with a dark theme. At the top, a tab is labeled 'index.html' with a close button. Below the tab, a breadcrumb navigation shows 'index.html > html > body'. The main code area displays an HTML document structure with line numbers 1 through 14 on the left. The code is as follows:

```
1  <html>
2
3  <head>
4    <title>Page Title</title>
5  </head>
6
7  <body>
8
9    <h1>This is a Heading</h1>
10   <p>This is a paragraph.</p>
11
12 </body>
13
14 </html>
```

HTML elements present in above code are 6 which are , `<html>` , `<head>` , `<title>`, `<body>`, `<h1>` and `<p>`.

Note : HTML tags are not case sensitive: `<P>` means the same as `<p>`.

## **HTML Attributes**

HTML attributes provide additional information about HTML elements.

Attributes usually come in name/value pairs like: name="value"

Eg :

```
<a href="https://www.google.com">Visit Google</a>



<p style="color: red;">This is a red paragraph.</p>
```

- All HTML elements can have attributes
- The `href` attribute of `<a>` specifies the URL of the page the link goes to
- The `src` attribute of `<img>` specifies the path to the image to be displayed
- The `alt` attribute of `<img>` provides an alternate text for an image
- The `style` attribute is used to add styles to an element, such as color, font, size, and more

## HTML headings and paragraph

HTML headings are defined with the `<h1>` to `<h6>` tags.

Eg

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

The HTML `<p>` element defines a paragraph.

A paragraph always starts on a new line, and browsers automatically add some white space (a margin) before and after a paragraph.

```
<p>This is a paragraph.</p>  
<p>This is another paragraph.</p>
```

## HTML formatting

Formatting elements were designed to display special types of text:

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

The HTML `<b>` element defines bold text, without any extra importance.

The HTML `<i>` element defines a part of text in an alternate voice or mood. The content inside is typically displayed in italic.

The HTML `<em>` element defines emphasized text. The content inside is typically displayed in italic.

The HTML `<small>` element defines smaller text:

The HTML `<mark>` element defines text that should be marked or highlighted:

The HTML `<del>` element defines text that has been deleted from a document. Browsers will usually strike a line through deleted text:

The HTML `<ins>` element defines a text that has been inserted into a document. Browsers will usually underline inserted text:

The HTML `<sub>` element defines subscript text. Subscript text appears half a character below the normal line, and is sometimes rendered in a smaller font. Subscript text can be used for chemical formulas, like H<sub>2</sub>O:

The HTML `<sup>` element defines superscript text. Superscript text appears half a character above the normal line, and is sometimes rendered in a smaller font. Superscript text can be used for footnotes, like WWW<sup>[1]</sup>:

Code example

```
<> index.html > html
1  <html>
2
3  <body>
4
5      <h1>This is a example for HTML formatting tags</h1>
6
7      <b>This text is bold</b>
8
9      <strong>This text is important!</strong>
10
11     <i>This text is italic</i>
12
13     <em>This text is emphasized</em>
14
15     <small>This is some smaller text.</small>
16
17     <p>Do not forget to buy <mark>milk</mark> today.</p>
18
19     <p>My favorite color is <del>blue</del> red.</p>
20
21     <p>My favorite color is <del>blue</del> <ins>red</ins>.</p>
22
23     <p>This is <sub>subscripted</sub> text.</p>
24
25     <p>This is <sup>superscripted</sup> text.</p>
26
27 </body>
28
29 </html>
```

O/p of the above example

---

# This is a example for HTML formatting tags

**This text is bold** **This text is important!** *This text is italic* *This text is emphasized* This is some smaller text.

Do not forget to buy **milk** today.

My favorite color is ~~blue~~ red.

My favorite color is ~~blue~~ red.

This is <sub>subscripted</sub> text.

This is <sup>superscripted</sup> text.

## HTML comments

Note : HTML comments are not displayed in the browser, but they can help document your HTML source code.

Syntax for HTML comment

```
<!-- Write your comments here -->
```

## HTML tables

HTML tables allow web developers to arrange data into rows and columns.

Sometimes you want your cells to be table header cells. In those cases use the `<th>` tag instead of the `<td>` tag:

`th` stands for table header.

```
index.html > html > body > table
1  <html>
2    <style>
3      table, th, td {
4        border:1px solid black;
5      }
6    </style>
7  <body>
8
9    <h2>A basic HTML table</h2>
10   <table style="width:100%">
11     <tr>
12       <th>Company</th>
13       <th>Contact</th>
14       <th>Country</th>
15     </tr>
16     <tr>
17       <td>Google</td>
18       <td>1234</td>
19       <td>USA</td>
20     </tr>
21     <tr>
22       <td>Facebook</td>
23       <td>1234</td>
24       <td>UK</td>
25     </tr>
26   </table>
27
28 </body>
29 </html>
```

O/p

| A basic HTML table |         |         |
|--------------------|---------|---------|
| Company            | Contact | Country |
| Google             | 1234    | USA     |
| Facebook           | 1234    | UK      |

## HTML List



HTML lists allow web developers to group a set of related items in lists.

An unordered list starts with the `<ul>` tag. Each list item starts with the `<li>` tag.

The list items will be marked with bullets (small black circles) by default:

An ordered list starts with the `<ol>` tag. Each list item starts with the `<li>` tag.

The list items will be marked with numbers by default:

```
<> index.html > html > body > ul > li
1  <html>
2  <body>
3
4      <h2>A basic HTML list</h2>
5      <ul>
6          <li>Coffee</li>
7          <li>Tea</li>
8          <li>Milk</li>
9      </ul>
10
11     <ol>
12         <li>Coffee</li>
13         <li>Tea</li>
14         <li>Milk</li>
15     </ol>
16
17
18 </body>
19
20 </html>
```

Output:

## A basic HTML table

- Coffee
- Tea
- Milk

1. Coffee
2. Tea
3. Milk

## HTML div element

The `<div>` element is used as a container for other HTML elements.

You can have many `<div>` containers on the same page.

```
<> index.html > html > body > div > p
1  <html>
2  <body>
3
4  <h1>Multiple DIV Elements</h1>
5
6  <div style="background-color: orange;">
7    <h2>Kohli</h2>
8    <p>Kohli is an Indian international cricketer</p>
9    <p>Kohli is widely regarded as one of the greatest all-format batters.</p>
10 </div>
11
12 <div style="background-color: white;">
13   <h2>Dhoni</h2>
14   <p>Dhoni is an Indian international cricketer</p>
15   <p>Dhoni is the most successful Indian captain.</p>
16 </div>
17
18 <div style="background-color: green;">
19   <h2>Rohit Sharma</h2>
20   <p>Rohit is an Indian international cricketer.</p>
21   <p>Rohit one of the greatest opening batters of all time.</p>
22 </div>
23
24 <p>CSS styles are added to make it easier to separate the divs</p>
25
26 </body>
27 </html>
```

## HTML classes:

The HTML `class` attribute is used to specify a class for an HTML element.

Multiple HTML elements can share the same class.

The `class` attribute is often used to point to a class name in a style sheet. It can also be used by a JavaScript to access and manipulate elements with the specific class name.

```
<> index.html > html > head
1  <html>
2
3  <head>
4      <style>
5          .city {
6              background-color: blue;
7              color: white;
8              border: 2px solid black;
9              margin: 20px;
10             padding: 20px;
11         }
12     </style>
13 </head>
14
15 <body>
16
17     <div class="city">
18         <h2>London</h2>
19         <p>London is the capital of England.</p>
20     </div>
21
22     <div class="city">
23         <h2>Paris</h2>
24         <p>Paris is the capital of France.</p>
25     </div>
26
27 </body>
28
29 </html>
```

- The HTML `class` attribute specifies one or more class names for an element
- Classes are used by CSS and JavaScript to select and access specific elements
- The `class` attribute can be used on any HTML element
- The class name is case sensitive
- Different HTML elements can point to the same class name

## HTML id attribute

The HTML `id` attribute is used to specify a unique id for an HTML element.

You cannot have more than one element with the same id in an HTML document.

The `id` attribute is used to point to a specific style declaration in a style sheet. It is also used by JavaScript to access and manipulate the element with the specific id.

```
<> index.html > html > head > style > .city
1  <html>
2  <head>
3    <style>
4      /* Style the element with the id "myHeader" */
5      #myHeader {
6        background-color: lightblue;
7        color: black;
8        padding: 40px;
9        text-align: center;
10     }
11
12     /* Style all elements with the class name "city" */
13     .city {
14       background-color: tomato;
15       color: white;
16       padding: 10px;
17     }
18   </style>
19 </head>
20 <body>
21
22   <h2>Difference Between Class and ID</h2>
23   <p>A class name can be used by multiple HTML elements, while an id name must only be used by one HTML element within the page:</p>
24   <!-- An element with a unique id -->
25   <h1 id="myHeader">My Cities</h1>
26
27   <!-- Multiple elements with same class -->
28   <h2 class="city">London</h2>
29   <p>London is the capital of England.</p>
30
31   <h2 class="city">Paris</h2>
32   <p>Paris is the capital of France.</p>
33 </body>
34
35 </html>
```

## HTML Layout

Websites often display content in multiple columns (like a magazine or a newspaper).

# HTML Layout Elements

HTML has several semantic elements that define the different parts of a web page:



- `<header>` - Defines a header for a document or a section
- `<nav>` - Defines a set of navigation links
- `<section>` - Defines a section in a document
- `<article>` - Defines independent, self-contained content
- `<aside>` - Defines content aside from the content (like a sidebar)
- `<footer>` - Defines a footer for a document or a section
- `<details>` - Defines additional details that the user can open and close on demand
- `<summary>` - Defines a heading for the `<details>` element

## Basic layout



## HTML URL

Web browsers request pages from web servers by using a URL.

A Uniform Resource Locator (URL) is used to address a document (or other data) on the web.

```
scheme://prefix.domain:port/path/filename
```

- scheme - defines the type of Internet service (most common is http or https)
- prefix - defines a domain prefix (default for http is www)
- domain - defines the Internet domain name (like w3schools.com)
- port - defines the port number at the host (default for http is 80)
- path - defines a path at the server (If omitted: the root directory of the site)
- filename - defines the name of a document or resource

## **HTML forms**

An HTML form is used to collect user input. The user input is most often sent to a server for processing.

The `<form>` element is a container for different types of input elements, such as: text fields, checkboxes, radio buttons, submit buttons, etc.

Basic form example

```
<> index.html > html > body > form
1  <html>
2    <body>
3
4      <form>
5        <label for="fname">First name:</label><br>
6        <input type="text" id="fname" name="fname"><br>
7        <label for="lname">Last name:</label><br>
8        <input type="text" id="lname" name="lname">
9      </form>
10   </body>
11
12  </html>
```

This is how the HTML code above will be displayed in a browser:

First name:

Last name:

The `<input type="radio">` defines a radio button.

Radio buttons let a user select ONE of a limited number of choices.

```
<> index.html > html > body > form > p
1  <html>
2    <body>
3      <form>
4        <p>Choose your favorite Web language:</p>
5        <input type="radio" id="html" name="fav_language" value="HTML">
6        <label for="html">HTML</label><br>
7        <input type="radio" id="css" name="fav_language" value="CSS">
8        <label for="css">CSS</label><br>
9        <input type="radio" id="javascript" name="fav_language" value="JavaScript">
10       <label for="javascript">JavaScript</label>
11     </form>
12  </body>
13
14  </html>
```

This is how the HTML code above will be displayed in a browser:

Choose your favorite Web language:

- ☐ HTML
- ☐ CSS
- ☐ JavaScript

The `<input type="checkbox">` defines a checkbox.

Checkboxes let a user select ZERO or MORE options of a limited number of choices.



```

<> index.html > html > body > form
1  <html>
2  <body>
3  <form>
4  <input type="checkbox" id="vehicle1" name="vehicle1" value="Bike">
5  <label for="vehicle1"> I have a bike</label><br>
6  <input type="checkbox" id="vehicle2" name="vehicle2" value="Car">
7  <label for="vehicle2"> I have a car</label><br>
8  <input type="checkbox" id="vehicle3" name="vehicle3" value="Boat">
9  <label for="vehicle3"> I have a boat</label>
10 </form>
11 </body>
12
13 </html>

```

The `<input type="submit">` defines a button for submitting the form data to a form-handler.

The form-handler is typically a file on the server with a script for processing input data.

The form-handler is specified in the form's `action` attribute.

```

<> index.html > html > body > form
1  <html>
2  <body>
3  <form action="/action_page.php">
4  <label for="fname">First name:</label><br>
5  <input type="text" id="fname" name="fname" value="John"><br>
6  <label for="lname">Last name:</label><br>
7  <input type="text" id="lname" name="lname" value="Doe"><br><br>
8  <input type="submit" value="Submit">
9  </form>
10 </body>
11
12 </html>

```

This is how the HTML code above will be displayed in a browser:

First name:

Last name:

If you click the "Submit" button, the form-data will be sent to a page called "/action\_page.php".

Action\_page.php is for understanding purpose only , please ignore for now as this concept will be dealt in Javascript later

## **HTML form elements**

The HTML `<form>` element can contain one or more of the following form elements:

- `<input>`
- `<label>`
- `<select>`
- `<textarea>`
- `<button>`
- `<fieldset>`
- `<legend>`
- `<datalist>`
- `<output>`
- `<option>`
- `<optgroup>`

The `<select>` element defines a drop-down list:

```
<> index.html > html > body > form > select#cars > option
1  <html>
2
3  <body>
4
5      <h2>The select Element</h2>
6
7      <p>The select element defines a drop-down list:</p>
8
9      <form action="/action_page.php">
10         <label for="cars">Choose a car:</label>
11         <select id="cars" name="cars">
12             <option value="volvo">Volvo</option>
13             <option value="saab">Saab</option>
14             <option value="fiat">Fiat</option>
15             <option value="audi">Audi</option>
16         </select>
17         <input type="submit">
18     </form>
19
20 </body>
21
22 </html>
```

The `<option>` element defines an option that can be selected.

By default, the first item in the drop-down list is selected.

To define a pre-selected option, add the `selected` attribute to the option:

```
<> index.html > html
1  <html>
2  <body>
3
4      <h2>Pre-selected Option</h2>
5
6      <p>You can preselect an option with the selected attribute:</p>
7
8      <form action="/action_page.php">
9          <label for="cars">Choose a car:</label>
10         <select id="cars" name="cars">
11             <option value="volvo">Volvo</option>
12             <option value="saab">Saab</option>
13             <option value="fiat" selected>Fiat</option>
14             <option value="audi">Audi</option>
15         </select>
16         <input type="submit">
17     </form>
18
19 </body>
20 </html>
```

Visible Values:

Use the `size` attribute to specify the number of visible values:

```
<> index.html >  html
1  <html>
2  <body>
3
4  <h2>Visible Option Values</h2>
5
6  <p>Use the size attribute to specify the number of visible values.</p>
7
8  <form action="/action_page.php">
9      <label for="cars">Choose a car:</label>
10     <select id="cars" name="cars" size="3">
11         <option value="volvo">Volvo</option>
12         <option value="saab">Saab</option>
13         <option value="fiat">Fiat</option>
14         <option value="audi">Audi</option>
15     </select><br><br>
16     <input type="submit">
17 </form>
18
19 </body>
20 </html>
```

The `<textarea>` Element

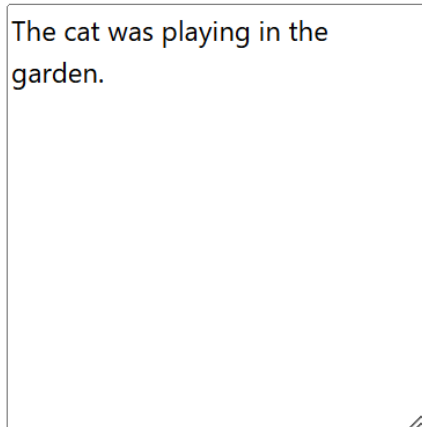
The `<textarea>` element defines a multi-line input field (a text area):

```
<> index.html >  html
1  <html>
2
3  <body>
4
5      <h2>Textarea</h2>
6      <p>The textarea element defines a multi-line input field.</p>
7
8      <form action="/action_page.php">
9          <textarea name="message" rows="10" cols="30">The cat was playing in the garden.</textarea>
10         <br><br>
11         <input type="submit">
12     </form>
13
14 </body>
15
16 </html>
```

The `rows` attribute specifies the visible number of lines in a text area.

The `cols` attribute specifies the visible width of a text area.

This is how the HTML code above will be displayed in a browser:

A screenshot of a web browser window. Inside the browser, there is a text area (textarea) containing the text "The cat was playing in the garden." The text area is rectangular with a thin border and a small scrollbar on the right side. The text is left-aligned and spans two lines.

The `<fieldset>` element is used to group related data in a form.

The `<legend>` element defines a caption for the `<fieldset>` element.

```
index.html > html > body > form
1  <html>
2  <body>
3
4  <h2>Grouping Form Data with Fieldset</h2>
5
6  <p>The fieldset element is used to group related data in a form, and the legend element defines a caption for the fieldset element.</p>
7
8  <form action="/action_page.php">
9    <fieldset>
10     <legend>Personalia:</legend>
11     <label for="fname">First name:</label><br>
12     <input type="text" id="fname" name="fname" value="John"><br>
13     <label for="lname">Last name:</label><br>
14     <input type="text" id="lname" name="lname" value="Doe"><br><br>
15     <input type="submit" value="Submit">
16   </fieldset>
17 </form>
18
19 </body>
20
21 </html>
```

# Grouping Form Data with Fieldset

The fieldset element is used to group related data in a form, and the legend element defines a caption for the fieldset element.

Personalia:

First name:

Last name:

HTML input types:

This chapter describes the different types for the HTML `<input>` element.

Here are the most commonly used input types

- `<input type="text">`
- `<input type="button">`
- `<input type="checkbox">`
- `<input type="email">`
- `<input type="file">`
- `<input type="number">`
- `<input type="password">`
- `<input type="submit">`

```

<> index.html > html > body > form > br
1  <html>
2  <body>
3
4      <form action="/action_page.php">
5          <label for="username">Username:</label><br>
6          <input type="text" id="username" name="username"><br>
7          <label for="pwd">Password:</label><br>
8          <input type="password" id="pwd" name="pwd"><br><br>
9          <input type="submit" value="Submit">
10     </form>
11
12 </body>
13
14 </html>

```

## HTML Links - hyperlinks

HTML links are hyperlinks.

You can click on a link and jump to another document.

```
<a href="https://www.google.com/">Open google</a>
```

The `target` attribute specifies where to open the linked document.

The `target` attribute can have one of the following values:

- `_self` - Default. Opens the document in the same window/tab as it was clicked
- `_blank` - Opens the document in a new window or tab
- `_parent` - Opens the document in the parent frame
- `_top` - Opens the document in the full body of the window

```
<a href="https://www.google.com/" target="_blank">Open google</a>
```

## HTML Encoding (Character Sets)

To display an HTML page correctly, a web browser must know which character set to use.

The character set is specified in the `<meta>` tag:

```
<head>
  <title>My Page</title>
  <meta charset="UTF-8">
</head>
```

## HTML Iframes

An HTML iframe is used to display a web page within a web page.

The HTML `<iframe>` tag specifies an inline frame.

An inline frame is used to embed another document within the current HTML document.

### Syntax

```
<iframe src="url" title="description"></iframe>
```

```
<iframe src="https://www.youtube.com/embed/HWGzQlrJOqM"></iframe>
```

## HTML 5

HTML5 is the fifth version of Hypertext Markup Language (HTML), a standard language used to structure webpages. It defines how content on a webpage should be structured and displayed. Here are some key points of HTML5

- Multimedia Support: Embeds audio and video without plugins.
- New Form Controls: Includes input types like date and email.



- Web Storage: Stores data offline for better performance.
- Semantic Elements: Uses tags like <header> and <footer> for better structure.
- Improved Performance: Faster and more efficient, especially on mobile.

## Sample HTML 5

```
1  <html>
2  <head>
3    <meta charset="UTF-8" />
4    <!--Responsive-->
5    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6    <title>HTML Page</title>
7  </head>
8  <body>
9    <header>
10     <h1>Hello, World!</h1>
11   </header>
12
13   <main>
14     <p>This is a simple HTML5 page.</p>
15   </main>
16
17   <footer>
18     <p>This is a paragraph</p>
19   </footer>
20 </body>
21 </html>
22
```

## Features of HTML5

HTML5 introduces several new elements and attributes to improve web development:

- **Semantic Elements:**
  - New tags like <article>, <aside>, <nav>, <header>, <footer>, and <section> provide meaning and structure to the content, improving both accessibility and SEO.
- **Multimedia Support:**
  - Native support for <audio> and <video> tags removes the need for third-party plugins like Flash.
- **Canvas API:**
  - Allows drawing graphics, animations, and games directly in the browser using JavaScript.
- **Geolocation API:**
  - Provides access to the user's geographical location (with permission), enabling location-based services.
- **Web Storage (localStorage and sessionStorage):**
  - Local storage (localStorage) and session storage (sessionStorage) provide client-side data storage, improving performance and enabling offline functionality.
- **Web Workers:**
  - Allows background processing of JavaScript, improving performance by preventing blocking of the main thread.
- **Form Enhancements:**
  - New input types (e.g., date, email, number) and form validation make forms easier and more efficient to create.



## HTML Canvas

The HTML <canvas> element is used to draw graphics on a web page.

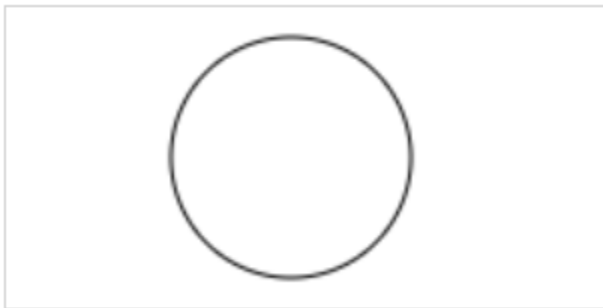
The HTML <canvas> element is used to draw graphics, on the fly, via JavaScript.

The <canvas> element is only a container for graphics. You must use JavaScript to actually draw the graphics.

Canvas has several methods for drawing paths, boxes, circles, text, and adding images.

Canvas is supported by all major browsers.

Here is an example of a basic, empty canvas:



```

<html>
<body>
  <p>Image to use:</p>
  

  <p>Canvas to fill:</p>
  <canvas
    id="myCanvas"
    width="250"
    height="300"
    style="border: 1px solid ■ #d3d3d3"
  >
    Your browser does not support the HTML canvas tag.</canvas>
  </>

  <p><button onclick="myCanvas()">Try it</button></p>

  <script>
    function myCanvas() {
      var c = document.getElementById("myCanvas");
      var ctx = c.getContext("2d");
      var img = document.getElementById("image");
      ctx.drawImage(img, 10, 10);
    }
  </script>
</body>
</html>

```

## HTML SVG Graphics

SVG (Scalable Vector Graphics)

[SVG defines vector-based graphics in XML](#), which can be directly embedded in HTML pages.

SVG graphics are scalable, and do not lose any quality if they are zoomed or resized:

```

<div class="svg-class">
  <?xml version="1.0" encoding="utf-8"?>

  <!-- Uploaded to: SVG Repo, www.svgrepo.com, Generator: SVG Repo Mixer Tools -->
  <svg
    fill="#000000"
    height="30px"
    width="30px"
    version="1.1"
    id="Layer_1"
    xmlns="http://www.w3.org/2000/svg"
    xmlns:xlink="http://www.w3.org/1999/xlink"
    viewBox="-271 273 256 256"
    xml:space="preserve"
  >
    <path
      d="M-64.5,273h-157c-27.3,0-49.5,22.2-49.5,49.5v52.3v104.8c0,27.3,22.2,49.5,49.5,49.5h157c27.3,0,49.5-22.2,49.5
v-52.3c-15.1,295.2-37.3,273-64.5,273z M-50.3,302.5h5.7v5.6v37.8l-43.3,0.1l-0.1-43.4l-50.3,302.5z M-179.6,374.7
c8.2-11.3,21.5-18.8,36.5-18.8s28.3,7.4,36.5,18.8c5.4,7.4,8.5,16.5,8.5,26.3c0,24.8-20.2,45.1-45.1,45.1s-44.9-20.3-44.9-45.1
C-188.1,391.2-184.9,382.1-179.6,374.7z M-40,479.5c-40,493-51,504-64.5,504h-157c-13.5,0-24.5-11-24.5-24.5v374.7h38.2
c-3.3,8.1-5.2,17-5.2,26.3c0,38.6,31.4,70,70,70c38.6,0,70-31.4,70-70c0-9.3-1.9-18.2-5.2-26.3h-40v479.5z"
    />
  </svg>
</div>

```

## HTML Video

The HTML `<video>` element is used to show a video on a web page.

```

<html>
  <body>
    <video width="320" height="240" controls>
      <source src="movie.mp4" type="video/mp4" />
    </video>
  </body>
</html>

```

The `controls` attribute adds video controls, like play, pause, and volume.

It is a good idea to always include `width` and `height` attributes. If height and width are not set, the page might flicker while the video loads.

The `<source>` element allows you to specify alternative video files which the browser may choose from. The browser will use the first recognized format.

The text between the `<video>` and `</video>` tags will only be displayed in browsers that do not support the `<video>` element.

## **HTML Audio**

The HTML `<audio>` element is used to play an audio file on a web page.

```
1  <html>
2    <body>
3      <audio controls>
4        <source src="horse.ogg" type="audio/ogg" />
5      </audio>
6    </body>
7  </html>
```

The `controls` attribute adds audio controls, like play, pause, and volume.

The `<source>` element allows you to specify alternative audio files which the browser may choose from. The browser will use the first recognized format.

The text between the `<audio>` and `</audio>` tags will only be displayed in browsers that do not support the `<audio>` element.

## **What is Web API?**

API stands for Application Programming Interface. An API is some kind of interface that includes a set of functions and subroutines that allow programmers to access specific features or data of an application, operating system or other services.

## **HTML APIs**

All browsers have a set of built-in Web APIs to support complex operations, and to help accessing data.

Here are some of the main HTML5 APIs:

1. [Geolocation API](#) - This API is used to access the current location of a user (with latitude and longitude).
2. [Drag and Drop API](#) - This API enables you to use drag-and-drop features in browsers.
3. [Web Storage API](#) - This API has mechanisms to let browsers store key/value pairs (in a more intuitive way than cookies).
4. [Web Workers API](#) - This API allows a JavaScript to run in the background, without affecting the performance of the page. Users can continue to do whatever they want: clicking, selecting things, etc., while the web worker runs in the background.
5. [Server-Sent Events API](#) - This API allows a web page to automatically get updates from a server.
6. [Canvas API](#) - This API lets you draw graphics, on the fly, via JavaScript.

Note : coding notes for HTML Web API will be shared along with Javascript post completion

